

The table-free latitudinal march

GMaster, session 36 – 12 September 2026

Lean 4 / Mathlib proof of record: `formal/lean/` (v4.34.0-rc2).

Abstract. We record the identities used by the table-free latitudinal march. For spin $s \geq 0$, order $m \geq 0$ and $x = \cos \theta$, the Wigner function is the Jacobi polynomial

$$d_{m,-s}^\ell(\theta) = (-1)^{m+s} 2^{\varepsilon_m \text{LG2N}(\ell,m)} \left(\sin \frac{\theta}{2}\right)^{m+s} \left(\cos \frac{\theta}{2}\right)^{|m-s|} P_{\ell-M}^{(m+s, |m-s|)}(x),$$

with $M = \max(m, s)$ and $\varepsilon_m = \text{sign}(m - s)$ (equal to $+1$ at $m = s$). The kernel marches this polynomial by the three-term recurrence, carries an exact binary exponent, emits a binade-split partial, and may skip polar tiles by ducc0’s m_{lim} cutoff. Every identity below is proved in Lean; only the WKB estimate of §8 and the IEEE scaling remark on the synthesis accumulator at the end of §5 are not.

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1 Introduction

The transform implemented by `gmaster/_spin_march_pallas.py` is, for each order m and each of four real channels $r_c(\theta)$,

$$A_c(\ell, m) = \sum_{\theta} d_{m,-s}^\ell(\theta) r_c(\theta), \quad \ell = \max(m, s), \dots, L - 1, \quad (1.1)$$

where $L = \ell_{\text{max}} + 1$. The kernel never stores associated Legendre tables. It evaluates $d_{m,-s}^\ell$ from a Jacobi recurrence in compensated float32, with an integer exponent lane, a float32 emit, and an optional polar skip.

This note is the human-readable statement. The proof of record is the Lean 4 project `GMasterMarch`; Appendix A names the theorems. The project has no `sorry`: DLMF 18.9.2 for the explicit Jacobi sum is proved in `JacobiSum.jacobi_three_term` (a telescoping certificate over the summands, §3), the identification of the Jacobi closed form with Wigner’s explicit sum in `WignerJacobi.wignerD_eq_jacobiForm` (§2), and the spin-2 mirror identity in `Mirror.mirror_identity` (§6). The WKB estimate in §8 is asymptotic and has no Lean counterpart.

The argument is organised as follows. §2 substitutes $m' \mapsto m$, $m \mapsto -s$ into Edmonds’s Jacobi form of the small- d matrix. §3 rearranges DLMF 18.9.2 into the marched coefficients and proves that $\alpha^2 - \beta^2 = 4ms$. §4–§5 treat the exponent lane and the binade-split emit. §6 records the hemisphere fold. §7 converts the live-degree count into a hardware lower bound. §8 identifies `ducc0`’s m_{lim} with the larger root of its quadratic.

2 Closed form

Write $\alpha = m + s$, $\beta = |m - s|$, $M = \max(m, s)$ and $n = \ell - M$.

Theorem 2.1.

$$d_{m,-s}^\ell(\theta) = (-1)^{m+s} 2^{\varepsilon_m \text{LG2N}(\ell,m)} \left(\sin \frac{\theta}{2}\right)^\alpha \left(\cos \frac{\theta}{2}\right)^\beta P_n^{(\alpha,\beta)}(x), \quad (2.1)$$

$$\text{LG2N}(\ell, m) = \log_2 \sqrt{\frac{(\ell+m)!(\ell-m)!}{(\ell-s)!(\ell+s)!}}, \quad \varepsilon_m = \begin{cases} +1, & m \geq s, \\ -1, & m < s. \end{cases} \quad (2.2)$$

Proof. Edmonds (1957, eq. 4.1.23) writes, for $d_{m'm}^j$ with $a = |m - m'|$, $b = |m + m'|$ and $n = j - (a + b)/2$,

$$d_{m'm}^j(\beta) = \xi \sqrt{\frac{n!(n+a+b)!}{(n+a)!(n+b)!}} \left(\sin \frac{\beta}{2}\right)^a \left(\cos \frac{\beta}{2}\right)^b P_n^{(a,b)}(\cos \beta), \quad (2.3)$$

where $\xi = 1$ if $m \geq m'$ and $\xi = (-1)^{m-m'}$ otherwise.

Substitute $m' \mapsto m$ and $m \mapsto -s$. Then $a = \alpha$, $b = \beta$, and $a + b = 2M$, so $n = \ell - M$. Unless $m = s = 0$ one has $-s < m$, hence $\xi = (-1)^{-s-m} = (-1)^{m+s}$. The factorial ratio splits on the two branches of M :

- if $m \geq s$, then $n = \ell - m$, $n + a + b = \ell + m$, $n + a = \ell + s$, $n + b = \ell - s$, so the ratio is $2^{+\text{LG2N}}$,
- if $m < s$, then $n = \ell - s$, $n + a + b = \ell + s$, $n + a = \ell + m$, $n + b = \ell - m$, so the ratio is $2^{-\text{LG2N}}$.

This is the ε_m flip in (2.2). □

The kernel writes $(-1)^m$ for $(-1)^{m+s}$. The two agree for even spin (0 and 2, the only spins the march serves). A spin-1 or spin-3 march must restore $(-1)^s$. The closed form *is* Wigner’s function: with $d_{m,-s}^\ell$ written as Wigner’s explicit sum (Wigner 1931; Sakurai 3.8.33), both sides are sums over $0 \leq k \leq j - M$, the powers of $\sin \frac{\theta}{2}$ and $\cos \frac{\theta}{2}$ match term by term ($\frac{x-1}{2} = -\sin^2 \frac{\theta}{2}$, $\frac{x+1}{2} = \cos^2 \frac{\theta}{2}$), and the coefficients agree by $\binom{j+s}{j-m-k} \binom{j-s}{k} (j-s-k)! (j-m-k)! (k+m+s)! k! = (j+s)! (j-s)!$ for $m \geq s$ (and $(j+m)! (j-m)!$ for $m < s$), which is where the ε_m flip comes from. Numerically, spins 0–3, $\ell \leq 12$, four colatitudes: worst relative difference 6×10^{-13} .

Lean: `Indices.alpha_add_beta`, `alpha_sub_beta`, `alpha_sq_sub_beta_sq`, `jratio_of_ge`, `jratio_of_lt`, `sign_of_even_spin`, `sign_of_odd_spin`; `WignerJacobi.wignerD` (Wigner’s sum), `jacobiForm`, `coeff_ge`, `coeff_lt`, `wignerD_eq_jacobiForm`.

3 Recurrence

DLMF 18.9.2 for $P_n^{(\alpha, \beta)}$, with $S = 2n + \alpha + \beta$, is

$$2n(n + \alpha + \beta)(S - 2)P_n = (S - 1)[S(S - 2)x + (\alpha^2 - \beta^2)]P_{n-1} - 2(n + \alpha - 1)(n + \beta - 1)SP_{n-2}. \quad (3.1)$$

Write $\text{den} = 2n(n + \alpha + \beta)(S - 2)$. Then $v_\ell = (c_1x + c_0)v_{\ell-1} - c_bv_{\ell-2}$ with

$$c_1 = \frac{(S - 1)S(S - 2)}{\text{den}}, \quad c_0 = \frac{(S - 1)(\alpha^2 - \beta^2)}{\text{den}}, \quad c_b = \frac{2(n + \alpha - 1)(n + \beta - 1)S}{\text{den}}. \quad (3.2)$$

Lemma 3.1. $\alpha^2 - \beta^2 = 4ms$ for every $m, s \geq 0$.

Proof. $\alpha - \beta = 2 \min(m, s)$ and $\alpha + \beta = 2 \max(m, s)$. $\square$

Thus no coefficient in (3.2) is a difference of large numbers. A float32 (high, low) split therefore carries them to 2^{-48} .

Seeds: $P_0 = 1$, and DLMF 18.5.7 at $n = 1$ gives

$$P_1^{(\alpha, \beta)}(x) = \frac{\alpha + \beta + 2}{2}x + \frac{\alpha - \beta}{2}. \quad (3.3)$$

For $s = 2$ and $m \geq 2$ this is $(m + 1)x + 2$.

Lean: `JacobiSum.jacobi` (the DLMF 18.5.7 sum), `tj` (its k -th summand in $u = \frac{x-1}{2}$), the ratio identities `tj_succ`, `tj_pred_v`, `tj_pred_u` and their combinations `tj_pred`, `tj_pred2`, the telescoping certificate `W` with `summand_telescopes`, `W_zero`, `W_top`, and DLMF 18.9.2 itself, `jacobi_three_term`; `Recurrence.jacobi_zero`, `jacobi_one`, `c0_eq`, `den_pos`, `recurrence_form`, `march`, `march_eq_jacobi`. No sorry in the chain.

Proof of DLMF 18.9.2 for the sum (as formalised). Let $t_k = \binom{n+\alpha}{n-k} \binom{n+\beta}{k} u^k (u+1)^{n-k}$ for $0 \leq k \leq n$ and $t_k = 0$ otherwise. From $\binom{N}{j+1}(j+1) = \binom{N}{j}(N-j)$ and $\binom{N}{j}(N+1) = \binom{N+1}{j}(N+1-j)$:

$$\begin{aligned} (\alpha + k + 1)(k + 1)(u + 1)t_{k+1} &= (n - k)(n + \beta - k)ut_k, \\ (n + 1 + \alpha)(n + 1 + \beta)(u + 1)t_k^n &= (n + 1 - k)(n + 1 + \beta - k)t_k^{n+1}, \\ (n + 1 + \alpha)(n + 1 + \beta)ut_k^n &= (\alpha + k + 1)(k + 1)t_{k+1}^{n+1}, \end{aligned}$$

and the difference of the last two expresses t_k^n through t_k^{n+1} , t_{k+1}^{n+1} with no division. Applying it twice writes the recurrence's summand at degree N , times $D' = (N + \alpha)(N + \beta)(N + \alpha - 1)(N + \beta - 1)$, through t_k, t_{k+1}, t_{k+2} alone, and it equals $W(k + 1) - W(k)$ for $W(k) = -q_0(k)t_k + (q_2(k - 1) - C(u + 1)(k + 1)(k + 1 + \alpha))t_{k+1}$ (explicit polynomials q_0, q_2, C , found with `sympy` and checked by `ring`); $W(0) = 0$ by the first identity at $k = 0$ and $W(N + 1) = 0$ because t vanishes past N , so the sum telescopes to zero.

4 Exponent carry

Every lane stores $v \cdot 2^{ex}$ with v float32 and $ex \in \mathbb{Z}$. Initially $v = 2^{f - \lfloor f \rfloor} \in [1, 2)$ and $ex = \lfloor f \rfloor$, where

$$f = \alpha \log_2 \sin \frac{\theta}{2} + \beta \log_2 \cos \frac{\theta}{2}. \quad (4.1)$$

Whenever $\max(|v_\ell|, |v_{\ell-1}|)$ leaves $[2^{-24}, 2^{24}]$, both lanes are multiplied by $2^{\mp 24}$ and ex moves by ± 24 .

Lemma 4.1. *The rescaling is exact and commutes with the recurrence.*

Proof. Multiplying a binary float by a power of two changes only the exponent field, hence is exact in the absence of overflow or underflow. With $|v| \leq 2^{24}$ times the growth of one step, and the float32 range $2^{\pm 126}$, overflow before the guard would require growth $> 2^{100}$. From (3.2) one has $|c_1|, |c_0|, |c_b| = O(m)$, so growth per step is $\leq 2^{15}$ at $m < 2^{14}$. The recurrence is linear and homogeneous in $(v_\ell, v_{\ell-1})$, so a common scale persists. $\square$

The band 2^{-24} is chosen so that $v \cdot r$ with $|r|$ of class 2^{24} and a 256-lane sum stays inside float32.

Lean: `ExponentCarry.lane_rescale`, `guard_large`, `guard_small`, `step_homogeneous`, `rec2_scaled` / `march_eq_rec2` / `march_scaled` (the march from seeds scaled by λ is λ times the march at every degree), `no_overflow_before_guard`.

5 Emit

At degree ℓ the program holds (v_θ, ex_θ) on 256 lanes and reduces $\sum_\theta v_\theta 2^{ex_\theta - e_{\max}} r_c(\theta)$ with $e_{\max} = \max_\theta ex_\theta$. Lanes with $ex_\theta - e_{\max} < -126$ are flushed to zero.

Lemma 5.1 (flush bound). *Flushed lanes change the tile partial by less than 2^{-70} relative to the largest lane's term.*

Proof. Stored $|v|$ lie in $[2^{-24}, 2^{24}]$. The largest term is at least $2^{-24} \cdot 2^{e_{\max}} \cdot |r|$; a flushed term is at most $2^{24} \cdot 2^{e_{\max} - 127} \cdot |r|$. The ratio is 2^{-79} , and there are at most 2^8 lanes. $\square$

From (2.1) the tile partial of (1.1) is

$$A_c^{\text{tile}}(\ell, m) = (-1)^{m+s} 2^{\varepsilon \text{LG2N} + e_{\max}} \sum_\theta [v_\theta 2^{ex_\theta - e_{\max}}] r_c(\theta). \quad (5.1)$$

Write $\varepsilon \text{LG2N} = \text{lex} + \text{frac}$ with $\text{lex} = \lfloor \varepsilon \text{LG2N} \rfloor$ and $\text{frac} \in [0, 1)$. The kernel stores

$$p = \text{fl}_{32} \left(\text{fl}_{32} \left(\sum_\theta \text{val}_\theta r_c(\theta) \right) \cdot \text{fl}_{32} \left((-1)^{m+s} 2^{\text{frac}} \right) \right), \quad e = \text{lex} + e_{\max}, \quad (5.2)$$

and the driver returns $\text{fl}_{64}(p) \cdot 2^e$, summing tiles in float64.

Proposition 5.2. $\text{fl}_{64}(p) 2^e$ equals (5.1) with relative error at most $(1+u)^2(1+\delta) - 1$, where $u = 2^{-24}$ and δ is the relative error of the float32 tree sum. Multiplication by 2^e is exact for $-1022 \leq e \leq 1023$. A partial with $e < -1022$ is below 2^{-1022} in magnitude and is dropped.

Proof. 2^e is exactly representable in float64 on that range, and multiplication by it only changes the exponent field. The two float32 roundings are $\text{fl}_{32}(2^{\text{frac}})$ and the product; each contributes a factor in $[1-u, 1+u]$. The tree error δ is the same term the previous float64-emit kernel carried. Harmonic coefficients are $O(1)$ sums of these partials, so a contribution below 2^{-1022} is below float64 resolution. $\square$

Measured relative change against the float64-emit kernel, over every lane: 7.5×10^{-8} , 7.3×10^{-8} , 6.6×10^{-8} at $N_{\text{side}} = 1024, 2048, 4096$, i.e. $\sim 2u$ as predicted.

Lean: `Emit.flushed_lane_small`, `flush_bound`, `binade_split`, `frac_factor_bounds`, `int_binade_is_zpov`, `two_roundings`, `dropped_partial_small`.

Synthesis accumulator (remark, not formalised). The synthesis kernel contracts the same marched row against harmonic coefficients and accumulates per lane. It used to carry a block exponent per accumulator, re-derived at every degree for each order sign. The coefficients of a window are now divided by the power of two at their maximum, so $|c| < 4$; with $|d_{m,-s}^\ell| \leq 1$ the accumulator over at most L degrees is below $2^{2+\log_2 L}$ in magnitude, inside float32 range with no exponent lane, and the alignment $n_d 2^t$ of the marched row is one exact power of two shared by both order signs (the mirror sign is ± 1). Multiplication by a power of two commutes with rounding, so the result is bit-identical to the block-exponent form except for terms below 2^{-126} of the window scale, which are flushed. The driver multiplies the result back by the window scale. Measured map error against ducc0 is unchanged to the printed digit (2.32×10^{-4} at $N_{\text{side}} = 1024$ spin 2, 8.19×10^{-4} at 2048 spin 0, maximum relative to the map maximum). This is a statement about IEEE scaling rather than an identity, and has no Lean counterpart.

6 Hemisphere fold

Lemma 6.1. $d_{m,0}^\ell(\pi - \theta) = (-1)^{\ell+m} d_{m,0}^\ell(\theta)$.

Proof. $d_{m,0}^\ell(\theta) = \sqrt{(\ell-m)!/(\ell+m)!} P_\ell^m(\cos \theta)$ and $P_\ell^m(-x) = (-1)^{\ell+m} P_\ell^m(x)$ (DLMF 14.7.17). Equivalently, $P_n^{(\alpha,\alpha)}(-x) = (-1)^n P_n^{(\alpha,\alpha)}(x)$ from the explicit sum, and $\alpha = \beta = m$ at spin 0. $\square$

HEALPix rings satisfy $\theta_{N-1-i} = \pi - \theta_i$ and share weight and ϕ_0 . The sum (1.1) therefore collapses, per northern lane, to

$$\sum_{\text{north}} d_i [G_i + (-1)^{\ell+m} G'_i]. \quad (6.1)$$

One recurrence, two right-hand sides; the parity sign is applied outside the kernel. This halves the lane count for spin 0.

Lean: `Fold.jacobi_symmetric_parity`, `row0_reflect`, `sum_pairs_even`, `sum_pairs_odd`, `fold_even`.

For spin 2 the kernel uses the mirror channel $T[m, \pi - \theta, \ell] = (-1)^{\ell+s} T[-m, \theta, \ell]$ (measured residual 2.3×10^{-12} over every row). It is the following identity.

Lemma 6.2 (mirror). *For $s \leq m \leq \ell$,*

$$d_{m,-s}^\ell(\pi - \theta) = (-1)^{\ell+m} d_{m,s}^\ell(\theta), \quad d_{-m,-s}^\ell(\theta) = (-1)^{m+s} d_{m,s}^\ell(\theta), \quad (6.2)$$

hence $d_{m,-s}^\ell(\pi - \theta) = (-1)^{\ell+s} d_{-m,-s}^\ell(\theta)$.

Proof. From the explicit sum, $P_n^{(\alpha,\beta)}(-x) = (-1)^n P_n^{(\beta,\alpha)}(x)$. Under $\theta \mapsto \pi - \theta$ one has $\sin \frac{\theta}{2} \leftrightarrow \cos \frac{\theta}{2}$ and $x \mapsto -x$, so the Jacobi form (2.1) of $d_{m,-s}^\ell$, with $(\alpha, \beta) = (m+s, m-s)$ and $n = \ell - m$, becomes $(-1)^{\ell-m}$ times the Jacobi form with $(\alpha, \beta) = (m-s, m+s)$. That is Edmonds's form (2.3) of $d_{m,s}^\ell$ (there $a = m-s$, $b = m+s$, the same n): the factorial ratio is symmetric in (a, b) and the sign ξ is $(-1)^{m+s}$ in both, so the first identity holds with $(-1)^{\ell-m} = (-1)^{\ell+m}$. The same form with $(m', m) = (-m, -s)$ has $a = m-s$, $b = m+s$ and $\xi = 1$, which is the second identity. Multiplying the two signs gives $(-1)^{\ell+m}(-1)^{m+s} = (-1)^{\ell+s}$. $\square$

Lean: `Mirror.jacobi_reflect`, `rowNeg_reflect` (the first identity on the Jacobi closed forms `rowNeg`, `rowPos`), `mirror_identity` (the kernel's form).

7 The arithmetic wall

Lemma 7.1. $\sum_{m < L} (L - m) = L(L+1)/2$.

Proof. Substitute $k = L - m$; the sum is $\sum_{k=1}^L k$. $\square$

Let N be the number of (m, ℓ, θ) triples visited. Before polar skip, $N = \frac{1}{2}L(L+1)\lceil n_\theta/2 \rceil$ on the folded spin-0 route and $N = \frac{1}{2}L(L+1)n_\theta$ at spin 2.

N_{side}	L	rings (spin 0 / 2)	N spin 0	N spin 2
1024	3072	2048 / 4095	9.7×10^9	1.9×10^{10}
2048	6144	4096 / 8191	7.7×10^{10}	1.5×10^{11}
4096	12288	8192 / 16383	6.2×10^{11}	1.2×10^{12}

An analysis pass with n_c channels costs at least $(2 + n_c)N$ fused multiply-adds. On this card the measured float64 FMA rate is $R_{64} = 0.45 \times 10^{12} \text{ s}^{-1}$ (datasheet 0.94×10^{12}). Hence any float64 SIMT spin-0 analysis at $N_{\text{side}} = 4096$ needs

$$T \geq \frac{6 \cdot 6.2 \times 10^{11}}{0.94 \times 10^{12}} \text{ s} = 4.0 \text{ s} \quad (7.1)$$

at the datasheet rate (8.3s measured), against ducc0's 2.02s on the host. No float64 kernel on this card can beat the CPU reference at the top of the board. This is why the march is a compensated float32 recurrence: its per-triple cost is about 30 float32 instructions against an issue rate of $\sim 6 \times 10^{13}$ lane-instructions per second, a floor of ~ 0.3 s per pass at $N_{\text{side}} = 4096$. After the binade-split emit of §5 the analysis kernel measures 22.5 ms per 128-order window at $N_{\text{side}} = 4096$ against a recurrence-only floor of 13.7 ms, so the emit is 39 per cent of the kernel and the compensated recurrence 61 per cent.

Lean: `Wall.live_degrees, time_lower_bound` ($cN \leq \text{fmas} \leq RT$ gives $T \geq cN/R$).

8 Polar skip

For $m > (\ell + \frac{1}{2}) \sin \theta$ the function $d_{m,-s}^\ell(\theta)$ is in its forbidden region. Writing $u = \sqrt{\sin \theta} d_{m,0}^\ell$,

$$u'' + \left[(\ell + \frac{1}{2})^2 - \frac{m^2 - \frac{1}{4}}{\sin^2 \theta} \right] u = 0. \quad (8.1)$$

With $\ell' = \ell + \frac{1}{2}$ the WKB action from the turning point θ_t ($\sin \theta_t = m/\ell'$) down to $\theta < \theta_t$ is

$$S = \int_\theta^{\theta_t} \sqrt{\frac{m^2}{\sin^2 t} - \ell'^2} dt \sim \frac{2\sqrt{2}}{3} \frac{\delta^{3/2}}{\sqrt{\ell' \sin \theta} \cos \theta}, \quad (8.2)$$

where $\delta = m - \ell' \sin \theta$. This estimate is not formalised.

ducc0 neglects, on the ring at θ , every order above

$$m_{\text{lim}} = s|\cos \theta| + \sqrt{(\ell_{\text{max}} \sin \theta + \text{ofs})^2 - s^2 \sin^2 \theta}, \quad \text{ofs} = \max(100, 0.01 \ell_{\text{max}}). \quad (8.3)$$

Lemma 8.1. (8.3) is the larger root of $m^2 - 2s|\cos \theta| m + s^2 - t_1^2 = 0$ on a ring with $\cos^2 \theta + \sin^2 \theta = 1$ and $t_1 = \ell_{\text{max}} \sin \theta + \text{ofs}$. For $s = 0$ one has $m_{\text{lim}} = \ell_{\text{max}} \sin \theta + \text{ofs}$.

Proof. The roots are $s|\cos \theta| \pm \sqrt{s^2 \cos^2 \theta - s^2 + t_1^2} = s|\cos \theta| \pm \sqrt{t_1^2 - s^2 \sin^2 \theta}$; the larger is (8.3). At $s = 0$ the square root is t_1 . $\square$

Skipping every order $m \geq M$ on one ring removes $(L - M)(L - M + 1)/2$ degrees (Lemma 7.1 applied to $L - M$).

Every $\ell \leq \ell_{\text{max}}$ then has $\delta \geq \text{ofs}$. At $\ell_{\text{max}} = 12287$, $\sin \theta = \frac{1}{2}$, one finds $S \sim 19$, so neglected values sit below $\sim 10^{-8}$ times the oscillatory amplitude. The march skips a tile when $m > \max_{\text{tile}} m_{\text{lim}}(\theta)$. Because the neglected set is the reference's own, agreement with ducc0 is unchanged to the printed digit.

N_{side}	analysis (tile 256)	synth. spin 0 (1024)	synth. spin 2 (512)
1024	13.9%	2.5%	9.3%
2048	17.5%	10.0%	14.8%
4096	19.3%	15.1%	17.9%

Lean: `PolarSkip.mlim_is_root, mlim_ge, mlim_spin_zero, skipped_degrees`.

A Lean correspondence

Build: `cd formal/lean && ~/.elan/bin/lake build`. There is no sorry; `lake env lean` `CheckAxioms.lean` prints the axioms of every main theorem (`propext`, `Classical.choice`, `Quot.sound`).

Section	File	Content
§2	<code>Indices.lean</code>	$\alpha + \beta = 2M$, $\alpha^2 - \beta^2 = 4ms$, both factorial branches, sign
§2	<code>WignerJacobi.lean</code>	Wigner’s explicit sum equals the Jacobi closed form, both branches
§3	<code>JacobiSum.lean</code>	Jacobi sum; DLMF 18.9.2 by summand ratios and a telescoping certificate
§3	<code>Recurrence.lean</code>	seeds, rearrangement, <code>march = jacobi</code>
§4	<code>ExponentCarry.lean</code>	exact rescaling, homogeneous step, no overflow before the guard
§5	<code>Emit.lean</code>	flush 2^{-71} , binade split, $[1, 2)$ mantissa, two roundings
§6	<code>Fold.lean</code>	$P_n^{(\alpha, \alpha)}(-x) = (-1)^n P_n^{(\alpha, \alpha)}(x)$, paired rings
§6	<code>Mirror.lean</code>	$P_n^{(\alpha, \beta)}(-x) = (-1)^n P_n^{(\beta, \alpha)}(x)$; the spin-2 mirror identity
§7	<code>Wall.lean</code>	Lemma 7.1; time bound from an FMA count and a rate
§8	<code>PolarSkip.lean</code>	Lemma 8.1; degrees removed by a cutoff

Not formalised: the WKB action (8.2), and the IEEE scaling remark on the synthesis accumulator (§5).

References

- [1] A. R. Edmonds, *Angular Momentum in Quantum Mechanics*, Princeton University Press, 1957.
- [2] NIST Digital Library of Mathematical Functions, <https://dlmf.nist.gov/>.